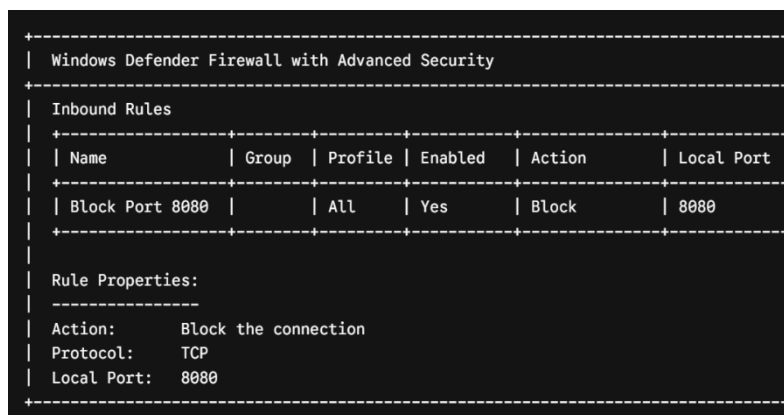


Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

Task 1 :- Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

ANS :-

1. Open Windows Defender Firewall with Advanced Security (search wf.msc in Start menu).
2. Click Inbound Rules on the left, then click New Rule... on the right.
3. Select Port → Click Next.
4. Choose TCP, enter 8080 under Specific local ports → Click Next.
5. Choose Block the connection → Click Next.
6. Check all profiles (Domain, Private, Public) → Click Next.
7. Name it Block Port 8080 and click Finish.



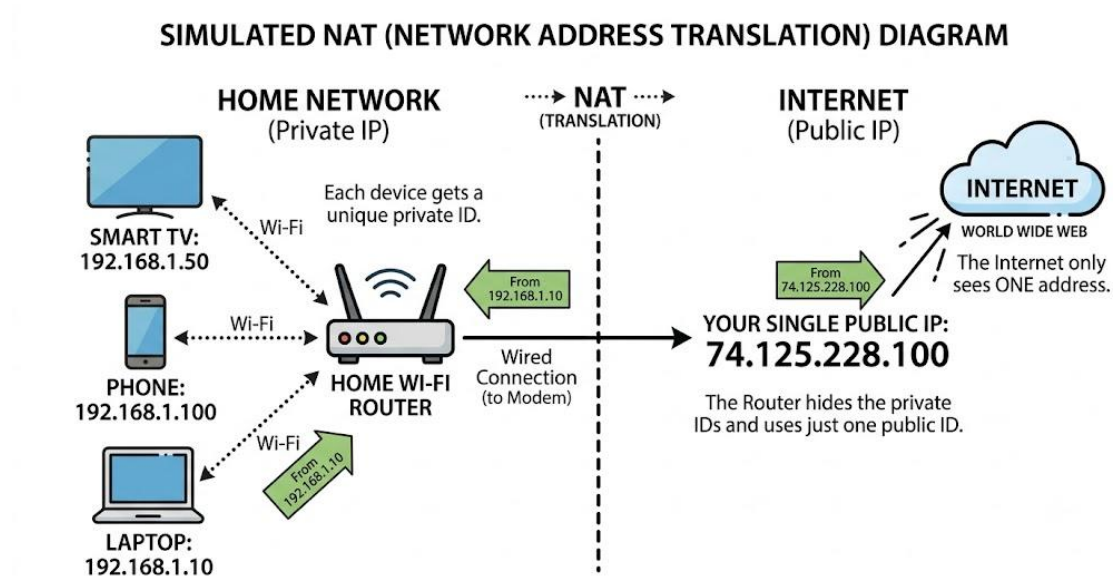
The screenshot displays the 'Windows Defender Firewall with Advanced Security' console. Under the 'Inbound Rules' section, a new rule named 'Block Port 8080' is visible. The rule is enabled and applies to all profiles. The action is set to 'Block the connection'. The protocol is 'TCP' and the local port is '8080'.

Name	Group	Profile	Enabled	Action	Local Port
Block Port 8080		All	Yes	Block	8080

Rule Properties:	
Action:	Block the connection
Protocol:	TCP
Local Port:	8080

Task 2 :- Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

ANS :-



Task 3 :- Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

ANS :-

Router Port Filtering Steps :-

1. Log into your router (192.168.1.1) and go to Security → Port Filtering (or Outbound Rules).
2. Click Add Rule, select protocol TCP, and set Port Range to 25.
3. Set action to Block / Reject, apply to All IP Addresses, and click Save/Apply.

How This Prevents Spam :-

Port 25 is used by email servers to send outgoing emails without encryption, which malware often exploits to send mass spam silently from infected home devices. Blocking this port stops hijacked laptops on your network from sending unauthorized spam emails to the outside world.

Task 4 :- Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

ANS :-

HTTP (192.168.1.5 → 172.217.166.78:80) : ALLOWED (*Standard web browsing port*).

SSH (10.0.0.10 → 34.201.23.1:22) : BLOCKED (*Secure shell port, usually blocked on public internet for security*).

DNS (192.168.1.7 → 8.8.8.8:53) : ALLOWED (*Google DNS port, essential for looking up websites*).

Task 5 :- Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

ANS :-

Blocking DDoS Cyberattacks :-

Firewalls inspect incoming traffic in real-time and block massive floods of fake junk packets (like TCP/UDP floods) so servers never crash under heavy load.

Rate Limiting & Bot Defense :-

Packet filters restrict how many requests a single IP address can make per second to block automated scripts from spamming APIs, scraping data, or credential stuffing.

Web Application Firewall (WAF) Filtering :-

Layer 7 WAFs read packet contents to identify and drop malicious payloads—such as SQL injections or cross-site scripting (XSS)—before they reach backend databases.